Software Analysis and Modeling Course

Software Analysis and Modeling Project

(*University Library Management System*)

Project Team:

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References:

Introduction

Academic libraries serve as the backbone of research and education within university environments. As digital transformation accelerates, it has become essential to develop robust systems that cater to the needs of modern users. This project focuses on designing the **University Library Management System (ULMS)** as a comprehensive technological solution to address existing challenges at Taibah University.

1.1 Problem Statement

The core computing problem addressed by this project is the fragmentation between manual library procedures and the need for a unified digital infrastructure. Currently, users face significant hurdles, including inconsistent real-time data regarding book availability on physical shelves and the necessity of on-site visits for basic administrative tasks. These operational bottlenecks lead to administrative delays and a suboptimal user experience.

1.2 Project Overview

The **ULMS** is a web-based platform designed to streamline the entire lifecycle of book circulation. It provides an interactive environment where students, staff, and faculty can manage their library activities remotely. The system ensures high security and reliability by utilizing existing University IDs and portal credentials for authentication and transaction verification.

1.3 Strategic Goals and Services

The proposed system aims to provide several high-level strategic services:

- **Enhanced Self-Service Experience:** Empowering users to independently manage their library needs—such as searching, digital holding, and reservation—through a 24/7 accessible web portal.
- **Operational Automation:** Reducing human intervention in inventory management and circulation by integrating barcode scanning technology for faster check-in and check-out processes.
- **Data Integrity and Transparency:** Providing accurate, real-time logging of all borrowing transactions to ensure precise inventory tracking and automated calculation of due dates and potential fines.

Current System

The current operational framework for library services at **Taibah University** utilizes a semi-automated infrastructure that relies on a combination of digital catalogs and manual on-site procedures. While the university provides digital access through its Online Public Access Catalog (OPAC) and the Saudi Digital Library (SDL), the ecosystem remains fragmented. This section analyzes the "As-is" state and compares it against local and international standards to justify the development of the **University Library Management System (ULMS)**.

2.1 Current Workflow and User Experience

Currently, stakeholders (Students, Staff, and Faculty) manage their library requirements through a workflow that faces several critical challenges:

- **Fragmented Access and Reliability:** Users frequently encounter accessibility issues, such as DNS resolution errors when attempting to access the library portal from outside the campus network. This limits the "High Availability" expected of a modern academic system.
- **Manual Inventory & Verification:** Although a digital catalog exists, real-time synchronization between physical shelf status and digital records is often inconsistent. Users often have to visit the library physically just to verify the availability of a specific copy.
- **Absence of Integrated Transactions:** The current system lacks a "Smart Basket" or a unified staging area for loan requests. Users cannot digitally "queue" books or define loan durations through a centralized web interface before arriving at the library.
- **Operational Bottlenecks:** For administration, the processes of check-in and check-out involve manual data entry or basic systems that are not fully integrated with a real-time web dashboard, leading to delays during peak hours.

2.2 Compares against local and international standards

To define the requirements for the proposed ULMS, we conducted a comparative analysis of industry-leading implementations:

A. Local & Regional Standards (Saudi Arabia):

- **King Saud University (KSU):** Employs the **Symphony** system, which features a robust "Self-Service" circulation module and automated notification systems—capabilities that are currently fragmented at Taibah University.
- **King Abdullah University of Science and Technology (KAUST):** Utilizes **Ex Libris Alma**, a cloud-based platform that offers "Unified Resource Management," ensuring a seamless experience across all digital and physical assets.

B. Global Standards (USA):

- **Harvard University:** Uses a highly integrated system that allows for "Request-to-Pickup" functionality and automated fee management through a digital student wallet.

- **Stanford University:** Utilizes the **FOLIO** platform, which supports extensible apps to manage diverse library workflows, serving as the blueprint for our proposed "Smart Basket" feature.

2.3 Rationale and Transition to the Proposed ULMS

The identified gap between Taibah's current fragmented state and the benchmarks at **KSU, KAUST, and Harvard** necessitates a comprehensive technological upgrade. The proposed **ULMS** is designed as a web-based solution that centralizes all functions using University and National IDs for authentication. It will resolve the current "Computing Problem" by introducing:

1. **Real-Time Inventory Management:** Ensuring 100% accuracy between the physical shelf and the digital dashboard.
2. **The "Smart Basket" System:** Enabling users to browse, select, and define loan durations remotely via a web portal.
3. **Automated Administrative Tools:** An advanced Admin Dashboard utilizing **Barcode Scanning technology** to automate the check-in/check-out process and synchronize inventory instantly.
4. **Integrated Financial Module:** Automating the calculation and tracking of loan fees and late penalties without manual administrative intervention.

Requirements elicitation

3.1 Methodology

We used scenario-based elicitation to extract the requirements needed by the Taibah Library System. This technique was useful in the way that it helped us go through how the system would work from a student's point of view. By narrating the scenario, the interactions between the university portal and the library databases became clear. This process helped in identifying the functional and non-functional requirements needed to develop the complete system model.

3.2 User Scenario

Khalid, a Computer Science student from Taibah University wanted to acquire a book from the library. Khalid visits the student portal where he can enter the library website to access the online catalog of books. He then searches for and finds the book he wanted in the main library. After that, Khalid clicks on "Reserve" which gets him the permission to borrow the book by placing it on hold for himself. The status of the book changes to "Reserved," preventing any other student from borrowing it. Khalid automatically receives a confirmation message through the university email address. The next day, he goes to the library and shows his student ID to the librarian to confirm his reservation and borrows the book completing the process.

3.3 System Requirements

Functional Requirements:

- The user must be able to log into the system using their Taibah University portal credentials.
- The user must be able to search for textbooks by title, author, or subject.
- The user must be able to place a digital hold on a book that is currently available in the library.
- The system must allow for the automatic delivery of confirmation emails to the user's university account once a reservation is finalized.
- The system must allow for the real-time tracking of book statuses, such as "Available," "Reserved," or "Checked Out."
- The system must allow the librarian to scan a student’s University ID and the book's barcode to complete a loan.
- The system must allow for the automatic creation of a loan record, including the calculation of a specific due date.
- The system must allow for the verification of student data through the university's Admissions and Registration Unit.

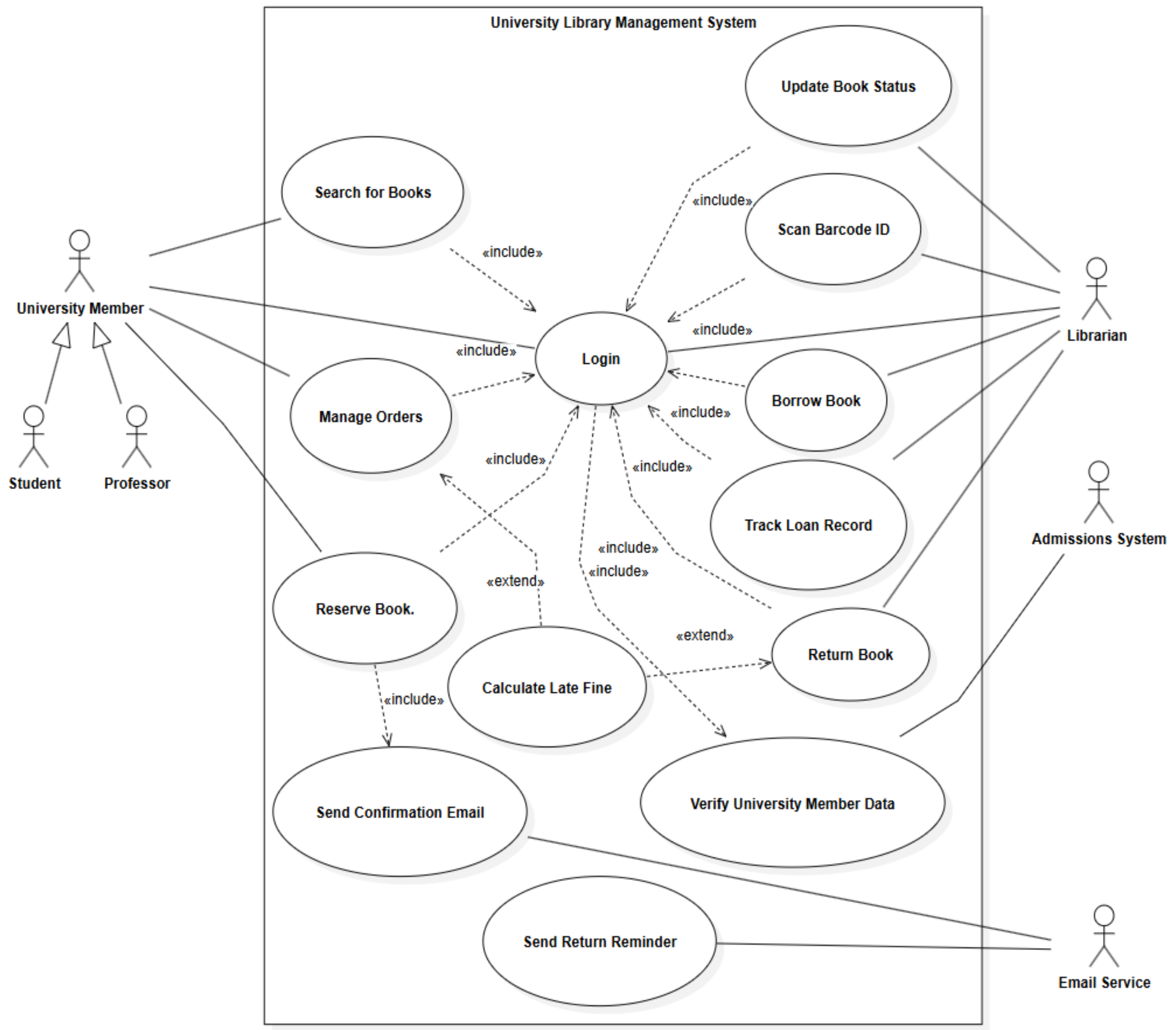
Non-Functional Requirements:

- The user must be able to navigate the search and reservation interface without requiring specialized training.
- The system must allow for the accurate logging of all transactions to prevent errors in student borrowing limits.
- The system must allow only authorized library staff to modify the status of a book from "Reserved" to "Checked Out."
- The user must be able to access the library catalog 24/7 through the electronic learning system.

System models

4.1 Functional/Use case Model

The Use Case Diagram below illustrates the major interactions between the primary actors—the **University Member** (comprising Students and Faculty) and the **Librarian**—and the University Library Management System (ULMS). Each actor is connected only to the operations relevant to their role, demonstrating a clear separation of privileges and administrative responsibilities. This diagram provides a high-level visualization of how the system supports user goals, such as the "Smart Basket" reservation system, and administrative control, including real-time inventory management and barcode-driven circulation.



The Use Case Diagram above demonstrates the core interactions between the system and its main actors, highlighting the transition from a fragmented infrastructure to a web-based solution. It clearly shows what the **University Member** does versus what the **Librarian** does, maintaining a well-organized and logical flow between use cases. The relationships between actions, such as **<<include>>** for mandatory portal authentication and barcode scanning, and **<<extend>>** for optional behaviors like late fee calculation, are used effectively to represent system dependencies. This makes the model both accurate for developers and easy to interpret for stakeholders.

Log in

System **University Library Management System**

Use case	Log In
Actors	University Member, Librarian
Description	Authenticates users to provide secure access to their respective dashboards and library services.
Stimulus	The user enters their university credentials on the login page.
Response	The system validates the credentials and redirects the user to the appropriate portal.
Scenario	"A student or staff member arrives at our portal and types in their official Taibah University ID and password. The system checks if they are who they say they are and then opens up their personal dashboard so they can start using the library services".
Comments	Only registered Taibah University members can access the system.

Search for Books

System **University Library Management System**

Use case	Search for Books
Actors	University Member
Description	Allows users to find specific textbooks or resources within the digital catalog.
Stimulus	The user enters search criteria such as title, author, or subject.
Response	The system displays a filtered list of books and their real-time availability.
Scenario	"When a student like Khalid needs a specific book for class, he goes to our website and simply types in the title or the author’s name. He should immediately see if the book is sitting on the shelf or if someone else has already taken it"
Comments	Accessible 24/7 through the student portal or learning system.

Manage Orders

System **University Library Management System**

Use case	Manage Orders
Actors	University Member
Description	Enables users to view and manage their active reservations and current loans.
Stimulus	The user opens the "My Orders" or "Smart Basket" section.
Response	The system displays a list of reserved and borrowed items with their due dates.
Scenario	"We want our users to have a 'Smart Basket' where they can see everything they’ve borrowed or reserved. It should be a central place where they can check their deadlines or cancel a hold on a book they no longer need".
Comments	This feature centralizes all user-specific transactions.

Reserve Book

System **University Library Management System**

Use case	Reserve Book
Actors	University Member
Description	Places a digital hold on a book, preventing others from borrowing it for a limited time.
Stimulus	The user clicks the "Reserve" button on an available book.
Response	The system changes the book status to "Reserved" and assigns it to the user.
Scenario	"If a student finds the book they want online, they should be able to click a 'Reserve' button. This should 'lock' the book for them so that nobody else can walk into the library and take it while they are on their way to pick it up".
Comments	The reservation ensures the book is ready for pickup the next day.

Borrow Book

System **University Library Management System**

Use case	Borrow Book
Actors	Librarian
Description	Finalizes the physical loan process and officially registers the book as borrowed.
Stimulus	The Librarian scans the student's ID and the book's barcode.
Response	The system creates a loan record and updates the book status to "Checked Out".
Scenario	"When a student comes to the front desk with a book, the librarian just needs to scan the student's ID card and the barcode on the back of the book. The system should then link that book to the student's account and set a return date automatically".
Comments	Real-time tracking prevents borrowing limit errors.

Return Book

System **University Library Management System**

Use case	Return Book
Actors	Librarian
Description	Processes returned books, closes loan records, and updates inventory.
Stimulus	The Librarian receives the book and scans its barcode.
Response	The system marks the book as "Available" and calculates any overdue fines.
Scenario	"When a book is brought back, the librarian scans it again. This should close the student's active loan and instantly update our system so that the very next person searching for that book sees it is 'Available' again"
Comments	Overdue returns will automatically trigger the "Calculate Late Fine" case.

Update Book Status

System **University Library Management System**

Use case	Update Book Status
Actors	Librarian
Description	Synchronizes physical library actions with the digital database in real-time.
Stimulus	A change in book availability occurs (e.g., a book is borrowed or returned).
Response	The system updates the digital dashboard status immediately.
Scenario	"We need the website to be a mirror of our actual library shelves. If a book is checked out at the desk, its status on the website should change to 'Checked Out' in that exact second so there is no confusion for other students".
Comments	Essential for maintaining 100% accuracy in inventory management.

Scan Barcode / ID

System **University Library Management System**

Use case	Scan Barcode / ID
Actors	Librarian
Description	Uses hardware scanning technology to automate data entry and speed up transactions.
Stimulus	The Librarian triggers the scanner for a book or university ID card.
Response	The system retrieves the unique identifier and fetches the associated record.
Scenario	"We want to move away from typing names and titles manually. We just want to point our barcode scanner at the student's card or the book, and have all the information pop up on the screen instantly to save time".
Comments	Reduces operational bottlenecks during peak library hours.

Track Loan Record

System **University Library Management System**

Use case	Track Loan Record
Actors	Librarian
Description	Provides an administrative overview of all loan activities and history.
Stimulus	The Librarian accesses the administrative dashboard reports.
Response	The system generates logs of all transactions and current borrowing limits.
Scenario	"As managers, we need to be able to pull up a full history of who has what. If a student is trying to take out too many books, the system should show us their current loan records so we can check if they've hit their limit".
Comments	Used for auditing and preventing student borrowing limit violations.

Verify University Member Data

System **University Library Management System**

Use case	Verify University Member Data
Actors	Admissions & Registration Unit
Description	Validates the academic standing and eligibility of a student or staff member.
Stimulus	A loan request is initiated by a Librarian.
Response	The system receives verification from the university's central database.
Scenario	"Before we hand over a book, our system should quickly talk to the University’s Registration Unit. We need to be sure the person standing at the desk is actually an active student or employee before we let them borrow anything".
Comments	Ensures only authorized university members utilize library resources.

Send Confirmation Email

System **University Library Management System**

Use case	Send Confirmation Email
Actors	Email Service
Description	Provides a digital receipt to the user for every successful transaction.
Stimulus	A reservation or loan is finalized in the system.
Response	The system sends an email to the user's university account.
Scenario	"Every time a student successfully reserves a book or checks one out, the system should shoot them an email automatically. This acts as their digital receipt so they know exactly what they have and when it needs to come back".
Comments	Essential for notifying users of their legal and financial obligations.

Calculate Late Fine

System **University Library Management System**

Use case	Calculate Late Fine
Actors	Librarian
Description	Automatically determines financial penalties for overdue book returns.
Stimulus	A book is scanned for return after its assigned due date.
Response	The system calculates the fine amount based on the number of overdue days.
Scenario	"If someone brings a book back late, we don't want to sit there with a calculator. The system should automatically see it's past the due date and tell us exactly how much the student owes in late fees".
Comments	This removes manual administrative intervention in financial tracking.

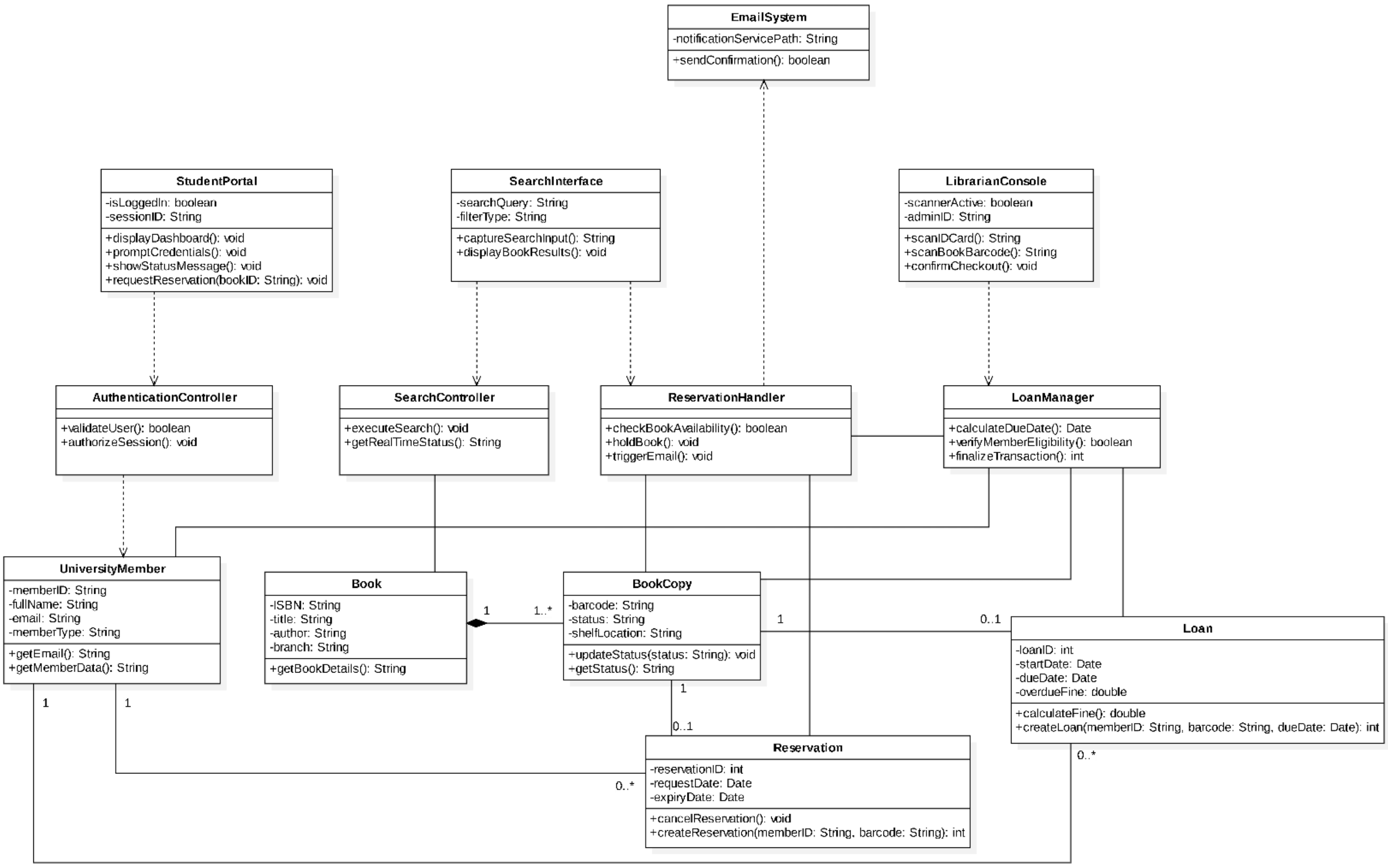
Send Return Reminder

System **University Library Management System**

Use case	Send Return Reminder
Actors	Email Service
Description	Proactively notifies users about upcoming deadlines to avoid late fines.
Stimulus	A system timer reaches a pre-defined threshold.
Response	The system sends a reminder email to the borrower.
Scenario	"To be fair to our students and help them avoid fines, the system should keep an eye on the calendar. A couple of days before a book is due, it should send a friendly reminder email to the student so they don't forget to bring it back".
Comments	Improves the "High Availability" and user experience of the system.

4.2 Object diagram Model

The class diagram below is an illustration of the static structure of the University Library Management System. The class diagram is a representation of all the components required to transition from fragmented manual procedures to a unified digital infrastructure. This model organizes the system into specialized layers to ensure the high security and reliability necessary for managing Taibah University's digital library activities.



The diagram illustrates the system through three primary layers:

User Interfaces: Positioned at the top, the classes StudentPortal, SearchInterface, and LibrarianConsole represent the system's "skin" and handle all the interactions with the *University Members* and the *Librarian*.

System Logic: The middle layer classes: AuthenticationController, SearchController, ReservationHandler, and LoanManager serve as the system's engine. They coordinate the automated workflows required to manage the book circulation, verify student eligibility, and trigger real-time email notifications.

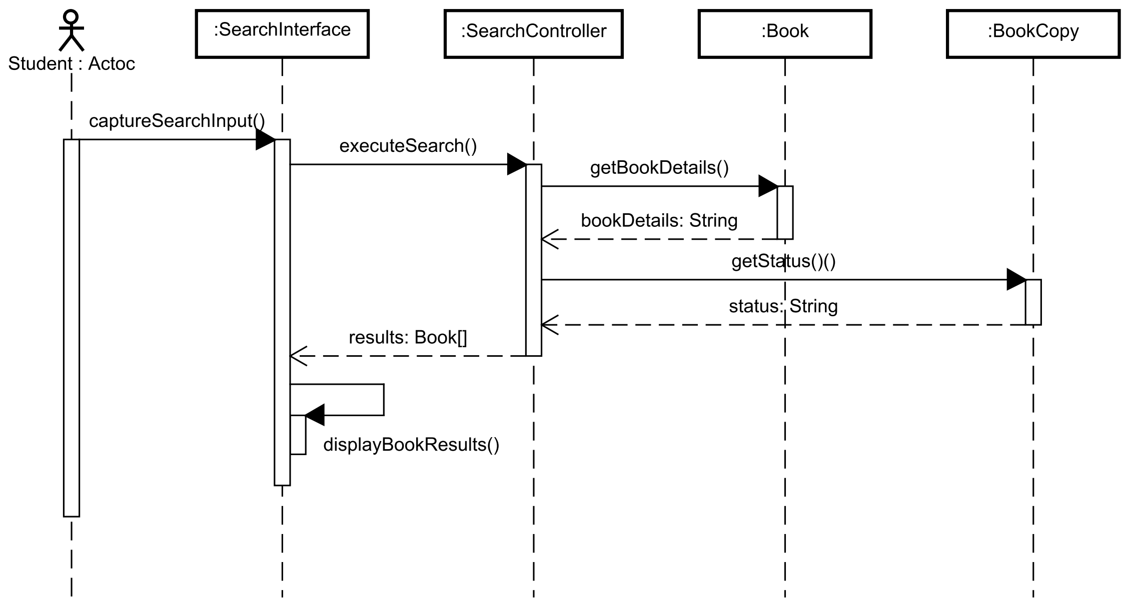
Data Entities and Integrity: The bottom layer classes: UniversityMember, Book, BookCopy, Reservation, and Loan represent the core of the system. They store all the library's information and use specific relationship rules to ensure the data is always accurate and the inventory is tracked in real-time.

4.3 Dynamic Model

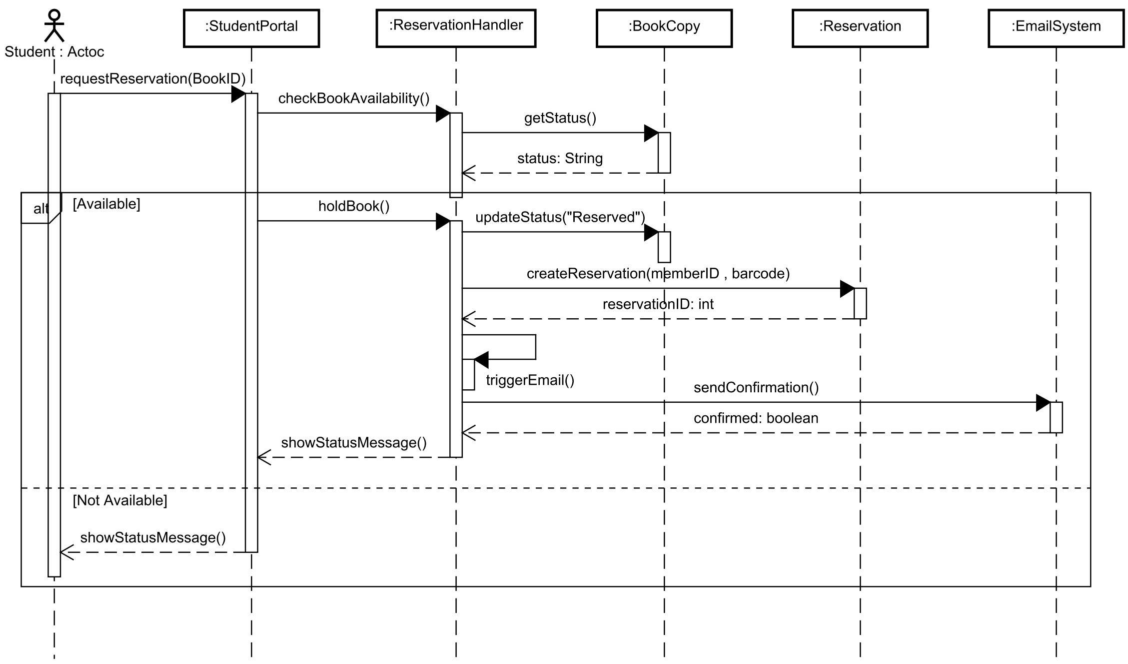
Sequence Diagram:

Following the static structure established in the Class Diagram, this section presents the **Dynamic Modeling** of the system. The following Sequence Diagrams illustrate how system objects interact with each other and with external Actors over time to fulfill the core functional requirements. Each diagram demonstrates the exchange of messages and method calls, ensuring 100% consistency with the previously defined Class Diagram. We have selected the three most critical scenarios within the University Library Management System (ULMS) for detailed representation:

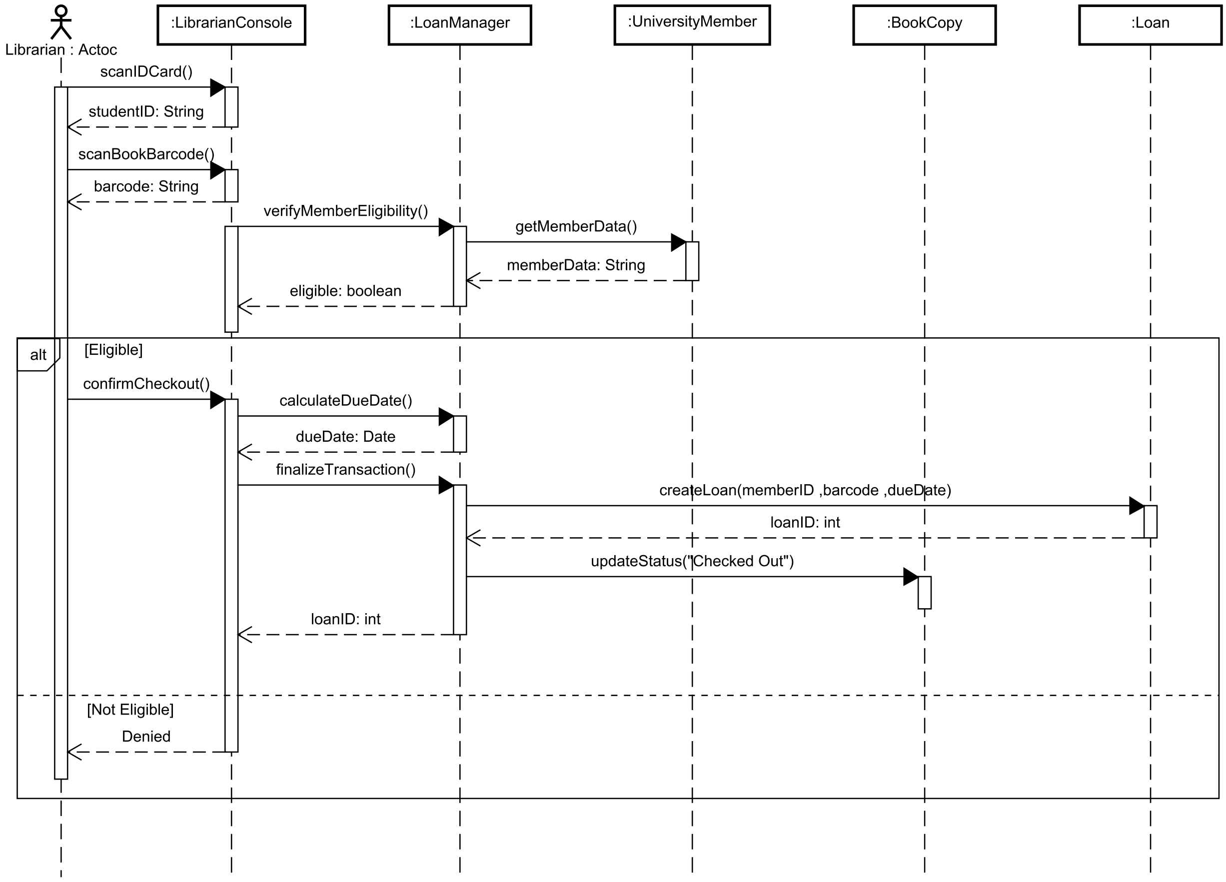
Search for Book:



Reserve Book:



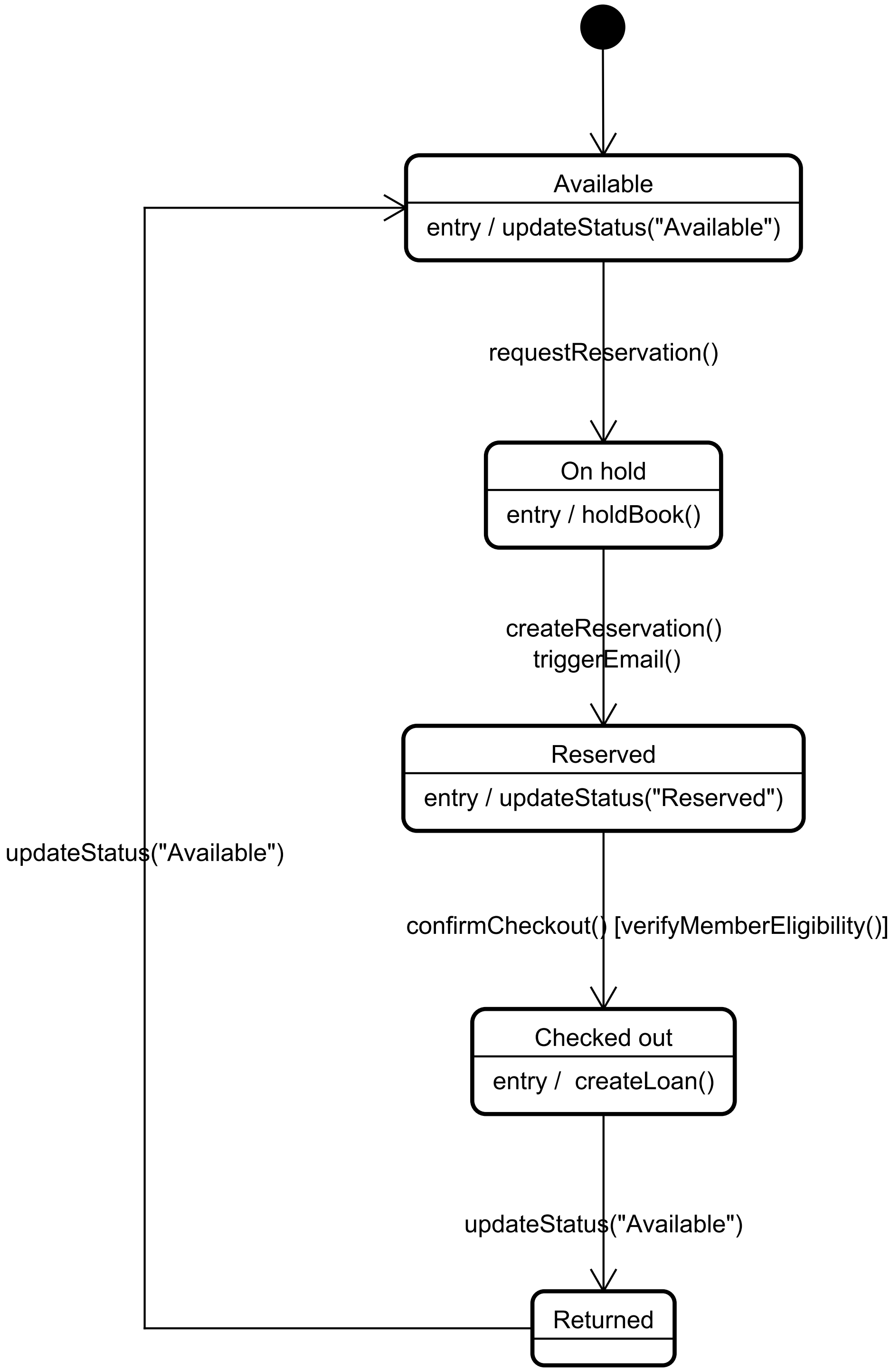
Borrow Book:



Librarian : Actoc
:LibrarianConsole
:LoanManager
:UniversityMember
:BookCopy
:Loan
scanIDCard()
studentID: String
scanBookBarcode()
barcode: String
verifyMemberEligibility()
getMemberData()
memberData: String
eligible: boolean
alt
[Eligible]
confirmCheckout()
calculateDueDate()
dueDate: Date
finalizeTransaction()
createLoan(memberID ,barcode ,dueDate)
loanID: int
updateStatus("Checked Out")
loanID: int
[Not Eligible]
Denied

4.4 State Machine Diagram:

Beyond time-based interactions, it is crucial to understand how the internal state of key system objects changes as a result of those interactions. This section presents the Object **Behavior Modeling** through a State Diagram for the BookCopy class, which is the most dynamic entity in the system. The diagram illustrates the lifecycle of a book copy, from its initial "Available" status on the shelf to "Reserved" or "Checked Out" states, and finally its return to the available pool. This representation ensures a clear understanding of all logical constraints and transitions applied to book data throughout various system operations.



Conclusion

5.1 Summary of Work

Throughout this project, we have successfully conducted a comprehensive **Requirements Analysis and Modeling** for the University Library Management System (**ULMS**). We transitioned from initial problem statements to formal **Functional and Non-functional Requirements**. By utilizing UML standards, we designed a robust **Class Diagram**, visualized dynamic interactions through **Sequence Diagrams**, and modeled complex object behavior with a **State Diagram** for the BookCopy entity. This systematic approach ensured that the system is not only functional but also architecturally sound and consistent.

5.2 Lessons Learned

- **Model Consistency:** One of the most vital lessons was ensuring that the static model (Class Diagram) and the dynamic models (Sequence/State) are perfectly synchronized. We learned that every method call in a sequence must have a corresponding definition in the class structure.
- **Encapsulation & Logic:** Designing the BookCopy state transitions taught us the importance of encapsulating state logic within the object itself to ensure data integrity.
- **Requirement Traceability:** We practiced how to trace a simple user scenario (like book reservation) through various layers of modeling to ensure the final design meets the user's needs.

5.3 Challenges Faced

- **Refining the Class Diagram:** The biggest challenge was the iterative process of refining the Class Diagram. We had to revisit and update our method signatures several times to accommodate the real-world logic discovered during the design of Sequence Diagrams.
- **Handling Conditional Logic:** Representing complex business rules (like eligibility checks and email triggers) within UML **alt fragments** required precise logical mapping to keep the diagrams clean yet comprehensive.

5.4 Possible Extensions

To further enhance the ULMS, the following extensions could be implemented:

- **Mobile Integration:** Developing a dedicated mobile application for students to scan barcodes directly from their phones for faster self-checkout.
- **AI-Powered Recommendations:** Integrating a machine learning module to suggest books to students based on their previous borrowing history and academic major.
- **Digital Resource Management:** Expanding the system to handle e-books, journals, and digital multimedia resources with a built-in PDF viewer and DRM protection.
- **Automated Notifications:** Implementing SMS and push notifications for overdue books and reservation availability to improve user engagement.